

Mini-Test
Modern Physics

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Text and solution for Monday's mini-test.

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WEEK III

- A hydrogen atom in excited state $n = 3$ decays to the ground state $n = 1$. Knowing that the energy of the ground state is $E_1 = -E_0 = -13.6 \text{ eV}$, calculate the energy of the initial state E_3 in electronvolts (1pt) and the wavelength λ of the emitted photon in nanometers (1pt). Give the results in the boxes with three significant digits. Use the constant $hc = 1240 \text{ eV nm}$.

Solution

The energy spectrum of the hydrogen atom ($Z = 1$) is given by

$$E_n = -Z^2 \frac{E_0}{n^2},$$

therefore

$$E_1 = -E_0 = -13.6 \text{ eV}$$

and

$$E_3 = -E_0/9 = -1.51 \text{ eV}.$$

The energy of the emitted photon is given by

$$E_\gamma = \Delta E_{3 \rightarrow 1} = 12.1 \text{ eV}$$

and the corresponding wavelength is

$$\lambda = \frac{hc}{E_\gamma} = 103 \text{ nm}$$

which is in the UV range, as expected for transitions in the Lyman series ($n = 1$).

Note that we used three significant digits and included the units for all answers.

- A hydrogen atom in excited state $n = 4$ decays to the ground state $n = 1$. Knowing that the energy of the ground state is $E_1 = -E_0 = -13.6 \text{ eV}$, calculate the energy of the initial state E_4 in electronvolts (1pt) and the wavelength λ of the emitted photon in nanometers (1pt). Give the results in the boxes with three significant digits. Use the constant $hc = 1240 \text{ eV nm}$.

Solution

Analogously to the other version, the results are

$$E_4 = -E_0/16 = -0.85 \text{ eV},$$

and

$$E_\gamma = \Delta E_{4 \rightarrow 1} = 12.75 \text{ eV}$$

and the corresponding wavelength is

$$\lambda = \frac{hc}{E_\gamma} = 97.3 \text{ nm}$$

which is in the UV range, as expected for transitions in the Lyman series ($n = 1$).

Note that we used three significant digits and included the units for all answers.